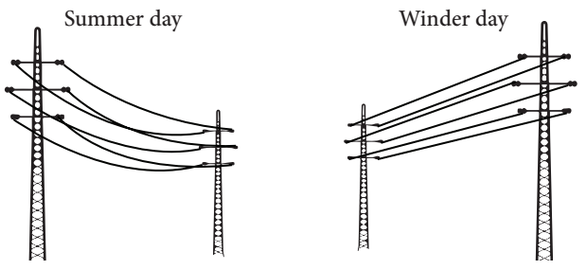


Glassware used in kitchen and laboratory are generally made up of Borosilicate glass (pyrex glass). The reason is that the Borosilicate glass do not expand much on being heated and therefore they do not crack.



so that they are free to change length.

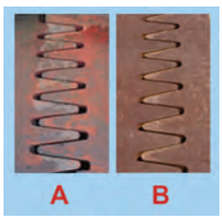


❖ The photographs below show an expansion joint at the end of a bridge in winter and in summer. Which season is shown in each picture? Explain how do you know?

_____

_____

_____



1.11 Numerical problems

1. I put a kettle containing 1 litre of cold water on the gas stove, and it takes 5 minutes to reach the boiling point. My friend puts on a small electric kettle, containing ½ litre of cold water, and it takes 5 minutes to get up to boiling point. Which gives more heat in 5 minutes?

a. the gas supply; or

b. the electricity supply?

Can you say how many times as much?

2. One calorie heat energy is needed to raise the temperature of the water from 30°C to 31°C. How much heat energy is needed to raise the temperature of the water from 30°C to 35°C.

Points to remember

- ❖ The main source of heat is sun, we can obtain heat from combustion, friction, and electricity.
- ❖ Heat is an energy that raises the temperature of a thing by causing the molecules in that thing to move faster
- ❖ Heat is the total Kinetic energy of constituent particles of objects.
- ❖ SI unit of Heat is joule (J).
- ❖ The measurement of warmness or coldness of a substance is known as its temperature.
- ❖ SI unit of temperature is kelvin.
- ❖ Temperature determines the direction of flow of heat when two bodies are placed in contact.
- ❖ Two objects are said to be in thermal contact if they can affect each other's temperature.
- ❖ Thermal equilibrium exists when two objects in thermal contact no longer affect each other's temperature.
- ❖ Most substances expand when heated and contract when cooled. The expansion of a substance on heating is called the thermal expansion of that substance.
- ❖ A solid has a definite shape, so when a solid is heated, it expands in all directions i.e., in length, area and volume, all increase on heating.

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